

UNIT 1: FOUNDATIONS OF PYTHON PROGRAMMING

Python is one of the most widely used programming languages in the IT industry today. It is used in web development, data science, artificial intelligence, automation, cybersecurity, and many other domains. Despite being a senior-level course, this unit assumes learners may have little to no prior experience with Python and therefore focuses on building a strong, well-explained foundation.

This unit introduces Python not merely as a programming language but as a problem-solving tool. Students will learn how programs are written, how Python code is executed, and how basic elements such as variables, data types, input/output, and operators work together to form meaningful programs. Emphasis is placed on clarity, readability, and understanding how Python differs from other languages students may have encountered.

By the end of this unit, learners are expected to confidently write simple Python programs, understand how data flows through a program, and prepare themselves for more complex topics such as control structures, functions, object-oriented programming, and web frameworks in later units.

INTENDED LEARNING OUTCOMES (ILOS)

At the end of this unit, students should be able to:

1. Explain fundamental programming concepts and Python's role in modern computing.
2. Install and configure a Python development environment correctly.
3. Identify and use basic Python data types appropriately.
4. Declare and manipulate variables following Python conventions.
5. Perform basic input and output operations in Python programs.
6. Apply arithmetic and basic operators to solve computational problems.

UNIT 1 PRETEST

Directions: Choose the best answer for each question.

1. Which characteristic of Python best supports rapid application development?
 - A. Low-level memory control
 - B. Hardware dependency
 - C. Readable and concise syntax
 - D. Manual memory management
2. Python executes programs primarily using a:
 - A. Compiler
 - B. Assembler
 - C. Interpreter
 - D. Linker
3. Which data type is most appropriate for storing a student number that will not be used in arithmetic operations?
 - A. int
 - B. float
 - C. str
 - D. bool
4. Which statement about Python variables is TRUE?
 - A. Variables must be declared with a data type
 - B. Variables are dynamically typed
 - C. Variables cannot change values
 - D. Variables store only integers
5. What is the output of the following code?

```
print(type(10 / 2))
```

 - A. int
 - B. float
 - C. str
 - D. bool

6. Which function allows a program to receive input from the user?
 - A. scan()
 - B. read()
 - C. input()
 - D. get()
7. Which operator is used for exponentiation in Python?
 - A. ^
 - B. **
 - C. //
 - D. %
8. What is the result of the expression `7 % 4`?
 - A. 1
 - B. 3
 - C. 4
 - D. 7
9. Which variable name follows Python naming conventions?
 - A. total-score
 - B. total score
 - C. total_score
 - D. 1total
10. Why is explicit type conversion often required when using `input()`?
 - A. `input()` returns integers
 - B. `input()` returns floats
 - C. `input()` returns strings
 - D. `input()` returns booleans

LESSON 1: INTRODUCTION TO PROGRAMMING WITH PYTHON

ESTIMATED DURATION:

1.5 hours

LESSON LEARNING OUTCOMES

After completing this lesson, students should be able to:

- Explain what programming is and why it is important
- Describe Python's key features and use cases
- Write and execute a simple Python program

Programming is the process of creating instructions that a computer can follow to perform specific tasks. These instructions are written using programming languages, which allow humans to communicate with computers in a structured and logical way.

Python is a **high-level programming language**, meaning it is closer to human language than machine language. It emphasizes readability and simplicity, making it an excellent choice for beginners and professionals alike. Python is also **interpreted**, which means code is executed line by line rather than compiled all at once.

Python is widely used in:

- Web development (e.g., Django, Flask)
- Data analysis and machine learning
- Automation and scripting
- Software prototyping

Example 1: Your First Python Program

```
print("Hello, World!")
```

Explanation:

- `print()` is a built-in Python function used to display output.
- The text inside quotation marks is a string.
- When executed, the program outputs: Hello, World!

This simple program demonstrates how Python executes instructions sequentially.

LESSON 1 ASSESSMENT

MCQs:

1. Python is primarily known for its:
 - A. Complex syntax
 - B. Readability and simplicity
 - C. Hardware control
 - D. Low-level operations
2. Python executes code:
 - A. All at once
 - B. Line by line
 - C. Only after compilation
 - D. Only in browsers
3. Which function displays output?
 - A. output()
 - B. print()
 - C. show()
 - D. echo()

Coding Exercise:

- Write a Python program that prints your name, program, and specialization on separate lines.

LESSON 2: SETTING UP THE ENVIRONMENT

ESTIMATED DURATION:

1 hour

LESSON LEARNING OUTCOMES

After completing this lesson, students should be able to:

- Install Python correctly
- Use an IDE or text editor for Python programming
- Run Python programs successfully

Before writing Python programs, a proper development environment must be set up. This includes installing Python and choosing a suitable editor or Integrated Development Environment (IDE).

Python can be downloaded from the official website and installed on Windows, macOS, or Linux. After installation, Python programs can be written using:

- IDLE
- VS Code
- PyCharm

Running a Python file via terminal:

```
python hello.py
```

Example: Running Python Code

```
print("Python is ready!")
```

Running this confirms that Python is installed correctly.

LESSON 2 ASSESSMENT

MCQs:

1. Which tool is used to write and run Python code?
 - A. Compiler
 - B. IDE
 - C. Browser
 - D. Database
2. Python files use which extension?
 - A. .txt
 - B. .exe
 - C. .py
 - D. .docx
3. Which IDE is commonly used for Python?
 - A. Eclipse
 - B. VS Code
 - C. NetBeans
 - D. Android Studio

Coding Exercise:

- Write and run a Python program that prints a confirmation message: “Python environment successfully configured.”.

LESSON 3: INTRODUCTION TO DATA TYPES

ESTIMATED DURATION:

2 hours

LESSON LEARNING OUTCOMES

After completing this lesson, students should be able to:

- Identify basic Python data types
- Differentiate between numeric and non-numeric data

Data types define the kind of value a variable can store. Python automatically determines data types, making it a dynamically typed language.

Common data types include:

- int – whole numbers
- float – decimal numbers
- str – text
- bool – True or False

Example:

```
age = 21
height = 1.75
name = "Alex"
is_enrolled = True
print(type(age))
```

Each variable stores a different data type.

LESSON 3 ASSESSMENT

MCQs:

1. Which data type stores text?
 - ☐ A. int
 - ☐ B. float
 - ☐ C. str
 - ☐ D. bool
2. What is the data type of True?
 - ☐ A. str
 - ☐ B. int
 - ☐ C. bool
 - ☐ D. float
3. Which is a floating-point value?
 - ☐ A. 10
 - ☐ B. 3.14
 - ☐ C. "5"
 - ☐ D. False

Coding Exercise:

- Declare variables for name, age, GPA, and enrollment status, then print each value and its data type.

LESSON 4: INTRODUCTION TO VARIABLES

ESTIMATED DURATION:

1.5 hours

LESSON LEARNING OUTCOMES

After completing this lesson, students should be able to:

- Apply Python variable naming rules
- Demonstrate dynamic reassignment of variables
- Use variables in arithmetic expressions

Variables act as labels that reference data stored in memory. Python allows reassignment, meaning the same variable name can point to different values during program execution.

Example:

```
score = 85
score = score + 5
print(score)
```

The variable score is updated dynamically.

Variable naming rules:

- Must begin with a letter or underscore
- Must not contain spaces or special characters
- Are case-sensitive

LESSON 4 ASSESSMENT

MCQs:

1. Which is a valid variable name?
 - ☐ A. 1score
 - ☐ B. my-score
 - ☐ C. my_score
 - ☐ D. my score
2. Python variables are:
 - ☐ A. Statically typed
 - ☐ B. Dynamically typed
 - ☐ C. Fixed
 - ☐ D. Immutable
3. Variables store:
 - ☐ A. Instructions
 - ☐ B. Data values
 - ☐ C. Functions
 - ☐ D. Errors

Coding Exercise:

- Create a variable for salary and compute annual income.

LESSON 5: BASIC INPUT AND OUTPUT OPERATIONS

ESTIMATED DURATION:

2 hours

LESSON LEARNING OUTCOMES

After completing this lesson, students should be able to:

- Use input() and print()
- Convert user input to appropriate data types

The print() function displays output, while the input() function collects user input. Since input() always returns a string, type conversion is necessary when performing numeric operations.

Example:

```
name = input("Enter your name: ")
print("Hello," , name)
```

Type Conversion Example:

```
age = int(input("Enter your age: "))
print(age + 1)
```

LESSON 5 ASSESSMENT

MCQs:

1. Which function gets user input?
 - A. read()
 - B. input()
 - C. scan()
 - D. get()
2. What data type does input() return?
 - A. int
 - B. float
 - C. str
 - D. bool
3. Which function displays output?
 - A. echo()
 - B. print()
 - C. show()
 - D. write()

Coding Exercise:

- Ask for two numbers and display their sum.

LESSON 6: BASIC OPERATORS

ESTIMATED DURATION:

2 hours

LESSON LEARNING OUTCOMES

After completing this lesson, students should be able to:

- Apply arithmetic operators
- Understand operator precedence

Arithmetic operators allow programs to perform calculations. Python follows the PEMDAS order of operations.

Arithmetic operators include:

- + addition
- - subtraction
- * multiplication
- / division
- % modulo
- ** exponentiation

Example 1:

```
x = 10
y = 3
print(x + y)
print(x ** y)
```

Example 2:

```
result = 10 + 2 * 5
print(result)
```

LESSON 6 ASSESSMENT

MCQs:

1. Which operator computes the remainder?
 - ☐ A. /
 - ☐ B. *
 - ☐ C. %
 - ☐ D. **
2. What is the result of $2 ** 3$?
 - ☐ A. 6
 - ☐ B. 8
 - ☐ C. 9
 - ☐ D. 23
3. Which operator has the highest precedence?
 - ☐ A. +
 - ☐ B. *
 - ☐ C. **
 - ☐ D. -

Coding Exercise:

- Write a program that calculates the area of a rectangle using user input.

UNIT 1 PERFORMANCE TASK

TASK TITLE:

Basic Student Information Program

TASK DESCRIPTION:

Students will create a Python program that collects and displays basic student information using fundamental programming constructs.

TASK REQUIREMENTS:

- Prompt the user to input the following details:
 - Full Name
 - Age
 - Course
 - Year Level
- Use appropriate data types for each input
- Display the collected information in a clear, formatted output

SAMPLE OUTPUT:

```
Student Information Summary
-----
Name: Hans Santos
Age: 20
Course: BSIT
Year Level: 3rd Year
```

RUBRIC (10 POINTS):

- Correct use of variables and data types – 4 pts
- Proper input and output formatting – 3 pts
- Program correctness – 3 pts

UNIT 1 POSTTEST

PART A: MULTIPLE CHOICE QUESTIONS

1. Which Python feature best supports rapid prototyping?
 - A. Manual memory management
 - B. Readable syntax
 - C. Hardware-level access
 - D. Strict typing
2. Python code is executed using a:
 - A. Compiler
 - B. Interpreter
 - C. Linker
 - D. Assembler
3. What is the return type of input()?
 - A. int
 - B. float
 - C. str
 - D. bool
4. Which data type is immutable?
 - A. int
 - B. float
 - C. str
 - D. bool
5. Which operator performs integer division?
 - A. /
 - B. //
 - C. %
 - D. **

6. What is the result of $5 + 2 * 3$?
 - A. 21
 - B. 11
 - C. 17
 - D. 13
7. Which variable name follows Python conventions?
 - A. total-score
 - B. total score
 - C. total_score
 - D. 1total
8. Which function displays output?
 - A. input()
 - B. display()
 - C. print()
 - D. write()
9. Which operator evaluates the remainder of a division?
 - A. /
 - B. //
 - C. %
 - D. **
10. Why is Python considered dynamically typed?
 - A. Variables must be declared
 - B. Data types are inferred at runtime
 - C. It uses a compiler
 - D. Variables cannot change type
11. What is the correct way to convert user input to an integer?
 - A. `int(input())`
 - B. `input(int())`

C. `integer(input())`

D. `convert(input())`

12. Which IDE is commonly used for Python development?

A. PyCharm

B. Eclipse only

C. NetBeans only

D. Xcode

13. What does the `type()` function return?

A. Variable value

B. Data type

C. Memory address

D. Function name

14. What is the result of `2 ** 3`?

A. 6

B. 8

C. 9

D. 12

15. Which statement about variables is TRUE?

A. Variables are fixed in value

B. Variables require type declaration

C. Variables can be reassigned

D. Variables store only numbers

PART B: CODING PROBLEMS

1. Write a Python program that accepts a user's name and year level, then displays a formatted greeting.
2. Write a Python program that accepts two numbers and displays their sum, difference, product, and quotient.

PART C: ESSAY QUESTIONS

1. Explain how Python's dynamic typing affects program flexibility and potential runtime errors.

2. Discuss the importance of proper variable naming and data type selection in professional software development.
